

KARAGANDY, KazakhstanWMO: **353940**Lat: **49.671N**Lon: **73.334E**Elev: **538**StdP: **95.03**Time Zone: **6.00 (E06)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.1	-29.9	-36.9	0.1	-33.2	-33.1	0.2	-29.5	15.0	-5.9	13.7	-6.2	2.5	70	0.649

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.2	32.1	16.8	30.1	16.3	28.2	15.7	18.5	27.2	17.7	26.3	17.1	25.5	3.8	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.0	12.1	20.4	15.0	11.4	19.7	14.0	10.7	19.5	54.6	27.1	52.2	26.4	50.0	25.2	23.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.8	10.9	9.3	DB	-35.7	35.4	3.7	2.1	-38.3	36.9	-40.5	38.1	-42.5	39.3	-45.2	40.8	(4)
(5)			WB	-35.7	20.5	3.6	1.2	-38.3	21.4	-40.4	22.1	-42.4	22.8	-45.0	23.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.3	-14.8	-13.6	-5.7	6.0	12.6	18.0	19.1	17.7	11.5	4.4	-5.2	-12.0
	DBStd	13.68	7.23	7.07	6.95	5.80	4.66	3.87	3.46	4.21	5.05	5.30	7.52	7.23
	HDD10.0	3454	769	660	487	142	28	1	0	2	41	184	457	683
	HDD18.3	5654	1028	893	744	370	183	51	31	62	211	432	707	942
	CDD10.0	995	0	0	1	23	110	241	283	241	86	11	0	0
	CDD18.3	153	0	0	0	1	6	41	56	44	6	0	0	0
	CDH23.3	2383	0	0	0	17	148	639	738	685	153	3	0	0
	CDH26.7	802	0	0	0	2	32	208	266	248	45	0	0	0
Wind	WSAvg	3.9	4.0	4.2	4.6	4.1	4.1	3.6	3.3	3.3	3.3	3.7	4.0	4.2
Precipitation	PrecAvg	359	24	21	26	27	38	39	54	26	18	27	30	29
	PrecMax	506	44	50	61	75	99	102	126	67	37	85	64	78
	PrecMin	230	4	4	6	3	10	2	8	1	0	7	4	10
	PrecStd	75	11	11	15	17	21	30	33	20	10	18	15	14
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.0	2.1	14.3	25.8	30.2	33.2	34.8	34.2	31.2	23.2	12.7	2.3
		MCWB	0.4	0.8	7.8	12.4	15.3	17.2	18.1	17.1	15.2	10.5	6.9	1.0
	2%	DB	-0.4	0.1	8.8	21.9	26.9	31.0	31.9	31.9	27.3	19.6	9.4	0.7
		MCWB	-1.3	-0.9	5.2	11.2	14.2	16.7	17.4	16.5	13.2	9.4	5.6	-0.3
	5%	DB	-2.7	-1.8	5.2	19.0	24.8	29.1	29.7	29.5	24.6	16.6	6.2	-0.9
		MCWB	-3.4	-2.7	3.0	9.9	13.2	16.0	16.8	16.0	12.6	8.2	4.0	-2.0
	10%	DB	-4.7	-3.2	3.0	16.1	22.8	27.1	27.3	27.1	21.9	13.5	3.7	-2.7
		MCWB	-5.4	-4.1	1.4	8.6	12.4	15.6	16.3	15.3	11.6	6.9	1.8	-3.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.6	1.0	8.4	13.5	16.8	19.2	19.7	19.1	16.1	11.7	8.8	1.5
		MCDB	1.5	1.8	13.2	22.2	26.4	28.4	28.7	28.2	28.2	19.7	10.5	2.0
	2%	WB	-1.3	-0.9	5.2	11.9	15.2	18.2	18.7	17.8	14.5	10.1	6.0	-0.3
		MCDB	-0.6	0.0	8.3	20.5	23.5	27.2	27.0	27.2	24.8	17.7	9.1	0.3
	5%	WB	-3.3	-2.6	3.1	10.5	14.2	17.3	18.0	16.9	13.2	8.9	3.9	-1.9
		MCDB	-2.7	-1.9	5.0	17.3	22.3	26.0	26.1	26.7	22.7	15.4	6.5	-1.1
	10%	WB	-5.4	-4.2	1.5	9.1	13.2	16.5	17.3	16.1	12.0	7.4	1.8	-3.5
		MCDB	-4.7	-3.4	2.8	15.1	20.8	24.8	25.1	25.6	20.5	12.5	3.5	-2.7
Mean Daily Temperature Range	5% DB	MDBR	9.9	10.7	9.8	12.5	14.6	14.1	13.2	14.7	14.8	11.8	9.0	9.4
		MCDBR	7.5	9.3	10.7	17.5	18.2	17.6	17.3	17.9	19.1	18.1	11.4	8.3
		MCWBR	7.1	8.7	7.7	8.7	8.3	7.1	6.6	6.9	8.6	9.9	8.1	7.7
	5% WB	MCDBR	7.5	8.9	10.1	15.5	15.5	15.1	14.5	15.6	17.4	15.8	10.7	7.9
		MCWBR	7.3	8.5	7.4	8.3	7.7	6.9	6.2	6.8	8.4	9.1	8.1	7.6
Clear-Sky Solar Irradiance	taub	0.291	0.325	0.376	0.408	0.409	0.411	0.436	0.400	0.354	0.339	0.325	0.296	
	taud	2.067	1.998	1.986	2.131	2.190	2.250	2.186	2.271	2.352	2.303	2.149	2.065	
	Ebn at Noon	719	785	810	833	850	849	819	833	840	777	672	645	
	Edh at Noon	86	119	145	141	138	132	138	121	100	88	80	75	
All-Sky Solar Radiation	RadAvg	1.59	2.77	4.07	4.95	6.23	6.58	6.00	5.51	4.21	2.37	1.24	1.14	
	RadStd	0.22	0.13	0.31	0.38	0.53	0.45	0.42	0.39	0.30	0.32	0.11	0.15	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-104	-37
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-104	-37

Nomenclature: See separate page